

Design of Experiments: Rubber Band Projectile Distance Length

Muhammad Ullah, ullahm7, 400529222

IBEHS 3S03

April 17, 2026

Dr. Carol Walker Bassim

Phase 0 — Proposal:

Elastic-based launching systems have been used in real-world applications ranging from equipment to engineering design throughout history. Catapults were first used to launch projectiles over long distances by releasing stored energy which were invented in Greco-Roman civilization during the 4th century BC [1]. During this time, ancient engineers also developed the ballista, which was an oversized crossbow, and the onager, which was a torsion-based machine from twisted ropes and animal sinew to power single arm [2]. In medieval times, trebuchet was developed that used gravity to throw large projectiles to break through strong fortified walls [3]. With the advent of vulcanized rubber in 1839 by Charles Goodyear, handheld catapult, also known as slingshot, was invented for small hunting practices such as squirrels or rabbits [4]. In terms of recent biomedical engineering problem, drone delivery systems operating in remote regions of Africa, such as Zipline's operations in Rwanda and Ghana, rely on catapult-style bungee launchers to propel drones carrying medical supplies into flight [5]. This allows medical supplies to be sent out to remote places with fast and reliable delivery for people that are in dire need instead of going on land, which will take longer and requires a lot of money and effort.

The rubber projectile for the design of experiment is a simplified analog of an engineering optimization problem. The objective of this experiment is to maximize the candidate response of horizontal distance traveled, in centimeters, by a rubber band launched from a fixed point. The first factor is launching angle, set at 30° (-1) and 60° (+1), 15° over and under the 45° angle which is managed by using protractor indication on paper to orient the launch. The second factor is stretch length, defined as the stretch of the rubber band from its resting position, set at 3 cm (-1) and 6 cm (+1), controlled by markings drawn for both pulls of different angles. The third factor is the natural resting radius of the rubber band by measuring the diameter and dividing it by 2, set at 1.5 cm (-1) and 3.5 cm (+1). For measuring distance, the launch will be done at a single point constant for each experimentation and using tape measurement to record the closest end of the rubber band to the point of launch.

The hypothesis for each factor includes the stretch length, angle launch, and the radius of the initial band. The stretch length prior to the launch is hypothesized to be the main effect as the larger the stretch, the greater the potential energy stored, which will result in greater launch velocity. The 6 cm stretch will have a higher distance than the 3 cm stretch as a result. The other hypothesis is that the higher launch angle, which is 60° , will result in greater distance than the 30° launch as it has more airtime. The last hypothesis is that the shorter radius of rubber band, which is 1.5 cm, will result in a greater distance than the 3.5 cm as it requires more tension to pull, which results in greater distance when released.

The main factor in controlling all experiments is the environment in which the launch will take place. The launch is taking place in the walkway indoors to control nuisance factors that can influence the experiment. Additionally, the launch will be taken at a constant height (80 cm) and will land on the same surface. The type of launch will also be consistent with the person using their hands, more so both thumbs to facilitate the angle and stretch. The possible disturbances that can arise during the launch is human error releasing the band as not all release will always be 100% the same for each launch, bounce and/or slide once the rubber band lands, and deformation of the rubber band when recording the distance. From a safety perspective, rubber band launching poses minimal risk; however, bystanders will be kept clear of the launch path during all runs. The stretch length has been tested and shows no tension that can cause breakage to occur. If discomfort were to occur due to excessive stretching of the rubber using the thumb, take breaks during the experimentation. Lastly, always launch the rubber bands away from the person experimenting by releasing the rubber band from the thumb that is closest to the person so that tension projects away to the front.

Phase 1 — Screening:

Since the experiment is dealing with 3 factors, it is a 2^3 coded factorial design. The experiment will have 8 runs in which each experiment is randomized to control any confounders. For the experiment to take place, a makeshift protractor and stretch length are made on paper to align when launching the rubber bands [Figure 1]. There point that is marked in grey is where the thumb is aligned with. There are 2 lines (one 30° and the other 60°) to adjust for which angle is being launched. There are 4 markings for each line, 2 markings that are in blue are for the rubber band with the short radius (1.5 cm), the other 2 markings are in red are for rubber band with the large radius (3 cm). Each marking of each color is for either short stretches (3 cm) or large stretch (6 cm). The paper is then attached to the chair in which the height of the chair plus the arm is 80 cm. The experimental runs are then taken using the chair and the paper to launch the rubber bands down the walkway [Figure 2]. After the launch has been made, the measurement is then recorded using a tape measure [Figure 3] that goes from the grey point at which the launch has been made to the bottom of the rubber band closest to the chair. If a rubber band is to hit an object, finger gets in the way or hits the walls surrounding the walkway; the run is voided and retested again.

The runs are randomized and recorded in the morning [Table 1]. This figure represents before any blocking for the afternoon launches has been made. With the blocks introduced into the models, which are both the AM and PM time of the day, it gives a total of 16 runs for the experiment [Table 2]. The ols shows the coded regression model as $y = \beta_0 + \beta_{\text{Block}} + \beta_A A + \beta_B B + \beta_C C + \beta_{AB} AB + \beta_{AC} AC + \beta_{BC} BC + \beta_{ABC} ABC$, for the experiment, where A = angle, B = stretch, and C = radius, with block introduced. Using the data table for the blocks, an ols model is provided [Figure 4]. The regression model now shows the coded version with the experimentation value assessed as $\text{Distance} = 408.25 + 31.50\text{Block} + 5.75A + 84.13B - 17.63AB - 102.38C - 51.88AC - 2.75BC - 13.75ABC$. The highest magnitude being β_0 , the intercept of the model which shows the average distance travelled (408.25 cm) during the experiment. The next greatest coefficient in terms of magnitude is C = -102.37; that means the lower the radius of the rubber band, it has a large effect on launch distance. This means that every unit of radius decreased, increased the distance by 102.37 cm. Next is B = 84.13, since it is positive, that means the greater the stretch length or every unit increase of stretch, it also has a good effect on the launch distance which increased by 84.13 cm. Lastly, the coefficient of A, which has the value of 5.72, does not show much effect by angle (60° and 30°) on launch distance. The anova table shows the sum of squared error (SSE) and mean squared error (MSE) of each coefficient [Figure 5]. The block can remove 3969 variance, however, there is still a significant amount that is still left over during the experimentation which suggests that experimentation time may not be a significant nuisance variable. This still shows that the total residual still is left with 8781 that is still unknown.

The Pareto plot ranks effects of the factors from largest to smallest [Figure 6]. This is especially useful in a screening experiment because it visually identifies which factors and interactions have the strongest influence on distance. In the figure, factor C, which is the radius of the initial band, has the greatest absolute effect size on the response distance. This was proven by the ols that decreasing the radius per unit leads to a significant increase in distance launched. This is then followed by factor B, AC, AB, ABC, and lastly A. Next is the interaction plot between the factors. The factors are plotting in 2-way interaction between each other; A with B, B with C, and C with A [Figure 7-9]. The interaction between A with B and A with C is significant as the lines are not parallel to each other. However, the interaction between B with C is not significant as they are almost parallel. Stretch and initial radius of rubber band act independently with one another as it is factors before and during the experiment. Though, the angle is a factor that interacts with radius (due to gravity) and stretches during the launch of the experiment. From these plots, B (stretch length) and C (radius of initial rubber band) will move on as it has the most effect on distance length from initial launch point.

Phase 2 — Optimization with Surface Response Methodology (SRM):

Next, it is to optimize the 2 factors that were selected, which were B (stretch length) and C (radius of initial rubber band). Since the angle (A) main effect was negligible and its interaction with radius was attributable to the dropped factor, the model was refitted using only B, C, and their interaction. The resulting coded regression model is [Figure 10]: $y = 424.00 + 97.25B - 117.75C - 8.75BC$. The B:C interaction coefficient of -2.75 is small relative to the main effects, confirming the Phase 1 finding that stretch and radius act largely independently of each other. The change of beta B is 97.25 and the change of beta C is -117.75 .

The surface plot has been produced that confirms maximizing distance from a fixed point [Figure 11]. The surface plot shows that when decreasing the stretch and increasing the radius, it will lead to a decrease in distance launched. In vice versa, when increasing the stretch and decreasing the radius, it will lead to an increase in distance launched. This is also made clear using the contour plot as well [Figure 12]. The hot regions (or hot colors) are indicated when stretch is +1 (6 cm) and when radius is -1 (1.5 cm) which shows the place where distance is large.

This shows the directionality to optimize distance is when the radius is small and stretch is large. The baseline of the experiment for stretch is 4.50 cm and the baseline for radius is 2.50 cm. To find the change of one coded unit of stretch, the change would be beta C/ beta B which is equal to $-117.75/97.25 = -1.21$. Therefore, a change in 1 coded unit for stretch will result in a -1.217 coded unit in radius of initial band size. To change it back to the real-world units is using the $-1.217 = \text{radius} - 2.5/1.0 = 1.28$ cm in radius. To get the y value, or response of distance travelled by the factors introduced, then it would be $y = 424.00 + 97.25(1) - 117.75(-1.217) - 8.75(1)(-1.217)$, which is 675.2 cm in distance length. This gets the 6th experiment which allows for the minimal radius that can allow for the largest distance. The reason to stop at this step is because if a step was to increase in stretch but decrease in radius, it will result in $-2.434 = \text{radius} - 2.5/1 = 0.066$ cm radius, which is almost impossible to do if it was stretched to 7 cm without having to much tension on the rubber band for it to snap. Instead for the 6th experiment, what can be done is leaving the radius of the rubber band at its minimum (the current radius of coded units of -1.217) and increasing the stretch length without having the continuously decrease the radius at the same time. The next coded unit for the distance is +2, which is $2(1.5) + 4.5 = 7.5$ cm for real world unit. The distance that comes out from the coded units of both the radius and stretch length is $y = 424.00 + 97.25(2) - 117.75(-1.217) - 8.75(2)(-1.217) = 783.1$. From these two optimization steps, the maximum distance that can be travelled is 783.1 cm from the launch point [Table 3].

Phase 3 — Reflection and next steps:

The most significant limitation of this experiment was the human launch mechanism. Although both thumbs were used consistently to facilitate the angle and stretch, no two releases are identical as variation can occur during the launch. This variation can include how the rubber band is deformed, shaking of the hand during the pull from the tension, trying to align the angle whilst also being in a comfortable position, or moment of release with the slight pull down of the thumb. If the experiment were to be repeated, a mechanical jig with a fixed release trigger can be used to hold the rubber band at a precise stretch of length and angle, eliminating human variability as a source of error entirely. This would substantially reduce the residual mean square of 1254 by reducing the error at the source.

Regarding blocking, the AM and PM session block effect was estimated at 31.5 cm but was not statistically significant. In hindsight, the blocking was precautionary and justified given that fatigue that can be experienced when having to deal with continuous recordings and measurements of the rubber band's launch, but the nuisance effect turned out to be small. If the experiment were repeated, the block can be on different days rather than having it done in the morning and then right after in the afternoon. This will reduce any noise of fatigue and bias from any experimentation done on the same day.

For next steps of optimization if given more time and effort would be to confirm true optimum existence of optimization. The linear model underlying this path cannot confirm whether a true optimum exists in this region or whether distance continues to increase indefinitely as stretch grows. The most important next step would therefore be fitting a second-order model using a Central Composite Design (CCD) centered at the steepest ascent endpoint. A CCD would add axial points at $B = \pm 1.414$ and $C = \pm 1.414$ (in coded units) around the new center, along with several center point replicates, allowing quadratic terms B^2 and C^2 to be estimated. These quadratic terms would reveal whether the response surface has a true maximum or whether it simply continues to rise, which would indicate that physical constraints such as band breakage rather than the physics of projectile motion are the binding limit on performance.

References:

- [1] “The Catapult: A History | January 2008 (112.1) | American Journal of Archaeology,” *American Journal of Archaeology*, Mar. 24, 2011. <https://ajaonline.org/book-review/536/>
- [2] “Medieval Weapons: Medieval Ballista. Types, Facts and History,” *Medieval Britain*, Mar. 10, 2023. <https://medievalbritain.com/type/medieval-life/weapons/medieval-ballista/>
- [3] M. McDonough, “Trebuchet | weapon,” *Encyclopedia Britannica*, Apr. 11, 2023. <https://www.britannica.com/technology/trebuchet>
- [4] “Vulcanized Rubber | EBSCO,” *EBSCO Information Services, Inc.* | www.ebsco.com, 2023. <https://www.ebsco.com/research-starters/history/vulcanized-rubber>
- [5] E. Dzisi, M. Patterson, and S. Iddrisu, “Drones in healthcare logistics: Insights from healthcare professionals’ perspective on Zipline delivery services in Ghana,” *Dialogues in Health*, p. 100285, Feb. 2026, doi: <https://doi.org/10.1016/j.dialog.2026.100285>.

Appendix:

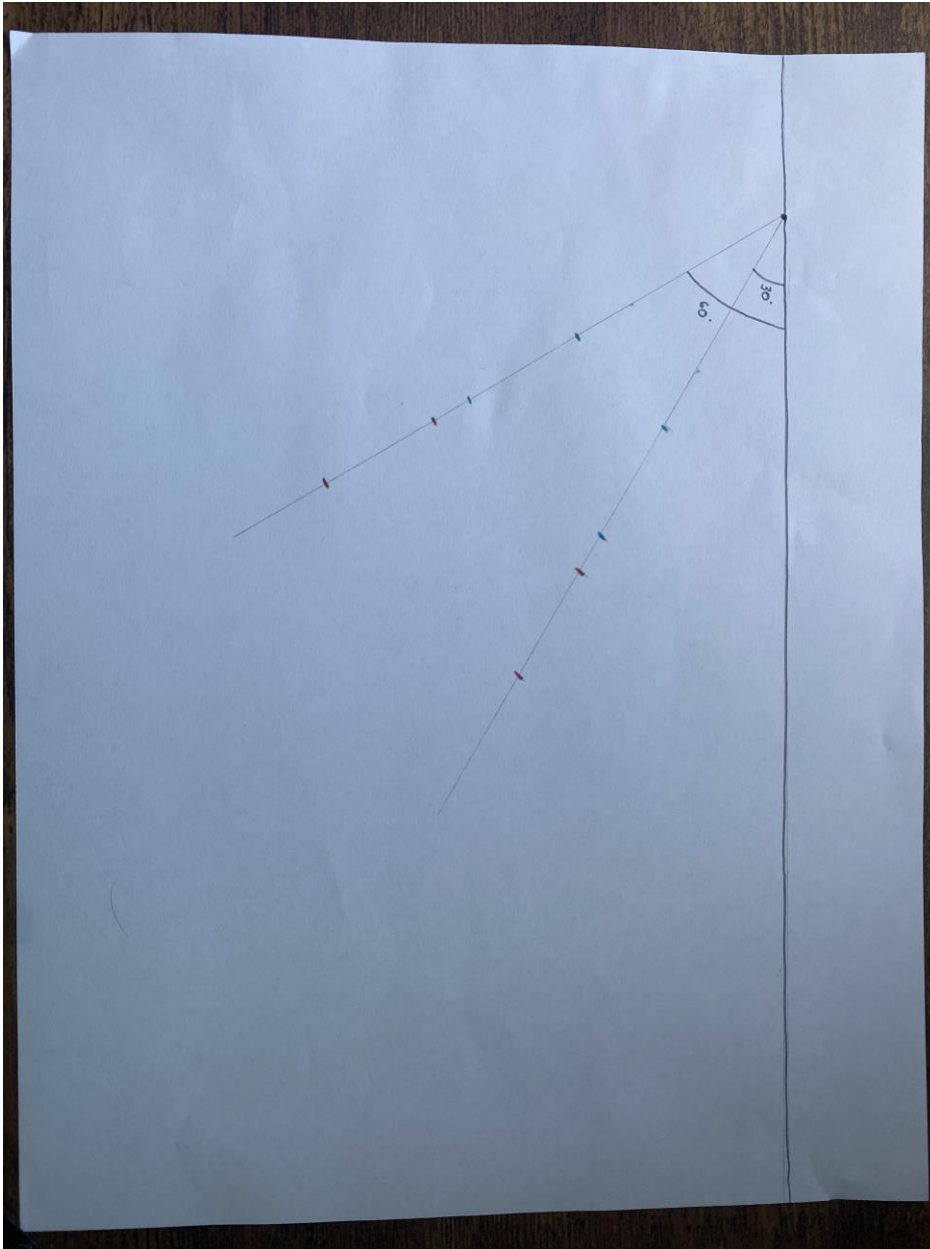


Figure 1: Protractor alignment with respective stretch length (red for 3.5 cm radius, blue for 1.5 cm radius)



Figure 2: Walkway of the experiment of rubber band launch



Figure 3: Material used for the rubber band launch

Table 1: Morning rubber band experimentation table with three factors

Experiment	Order	Angle	Stretch	Radius	Distance (cm)
1	6	-	-	-	354
2	5	+	-	-	486
3	2	-	+	-	562
4	8	+	+	-	702
5	3	-	-	+	209
6	1	+	-	+	195
7	7	-	+	+	448
8	4	+	+	+	310

Table 2: Morning and afternoon rubber band experimentation table with three factors and blocking

Experiment	Order	Angle (degree)	Stretch (cm)	Radius (cm)	Block	Distance (cm)
1	6	-	-	-	AM	354
2	5	+	-	-	AM	486
3	2	-	+	-	AM	562
4	8	+	+	-	AM	702
5	3	-	-	+	AM	209
6	1	+	-	+	AM	195
7	7	-	+	+	AM	448
8	4	+	+	+	AM	310
9	13	-	-	-	PM	402
10	9	+	-	-	PM	516
11	16	-	+	-	PM	557
12	12	+	+	-	PM	632
13	14	-	-	+	PM	301
14	11	+	-	+	PM	256
15	13	-	+	+	PM	513
16	10	+	+	+	PM	341

```

Results: Ordinary least squares
=====
Model:                OLS                Adj. R-squared:    0.946
Dependent Variable:   Distance            AIC:              164.3301
Date:                2026-04-13 19:44      BIC:              171.2834
No. Observations:    16                  Log-Likelihood:   -73.165
Df Model:             8                   F-statistic:      33.54
Df Residuals:        7                   Prob (F-statistic): 6.62e-05
R-squared:            0.975               Scale:           1254.4
=====
              Coef.   Std.Err.    t    P>|t|    [0.025   0.975]
-----
Intercept    408.2500   12.5221   32.6023  0.0000   378.6399  437.8601
Block[T.PM]   31.5000   17.7090    1.7788  0.1185   -10.3750   73.3750
A              5.7500    8.8545    0.6494  0.5368   -15.1875   26.6875
B             84.1250    8.8545    9.5008  0.0000    63.1875  105.0625
A:B          -17.6250    8.8545   -1.9905  0.0868   -38.5625    3.3125
C          -102.3750    8.8545  -11.5619  0.0000  -123.3125  -81.4375
A:C          -51.8750    8.8545   -5.8586  0.0006   -72.8125  -30.9375
B:C           -2.7500    8.8545   -0.3106  0.7652   -23.6875   18.1875
A:B:C        -13.7500    8.8545   -1.5529  0.1644   -34.6875    7.1875
=====
Omnibus:            0.789          Durbin-Watson:      1.952
Prob(Omnibus):      0.674          Jarque-Bera (JB):    0.027
Skew:               0.000          Prob(JB):           0.987
Kurtosis:           3.201          Condition No.:       3
=====
Notes:
[1] Standard Errors assume that the covariance matrix of the
errors is correctly specified.

```

Figure 4: OLS model of the morning and afternoon rubber band experimentation

	df	sum_sq	mean_sq	F	PR(>F)
Block	1.0	3969.00	3969.000000	3.163990	0.118509
A	1.0	529.00	529.000000	0.421706	0.536805
B	1.0	113232.25	113232.250000	90.266000	0.000030
A:B	1.0	4970.25	4970.250000	3.962163	0.086825
C	1.0	167690.25	167690.250000	133.678596	0.000008
A:C	1.0	43056.25	43056.250000	34.323397	0.000625
B:C	1.0	121.00	121.000000	0.096458	0.765165
A:B:C	1.0	3025.00	3025.000000	2.411457	0.164394
Residual	7.0	8781.00	1254.428571	NaN	NaN

Figure 5: ANOVA model of the morning and afternoon rubber band experimentation

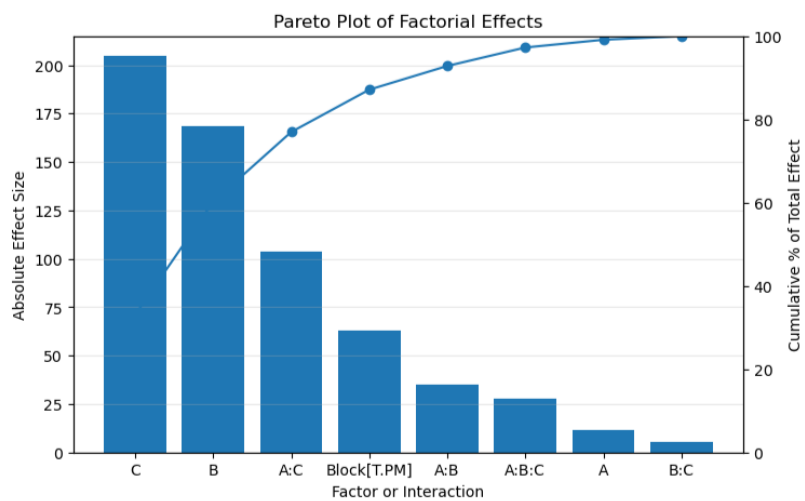


Figure 6: Pareto plot model of the morning and afternoon rubber band experimentation

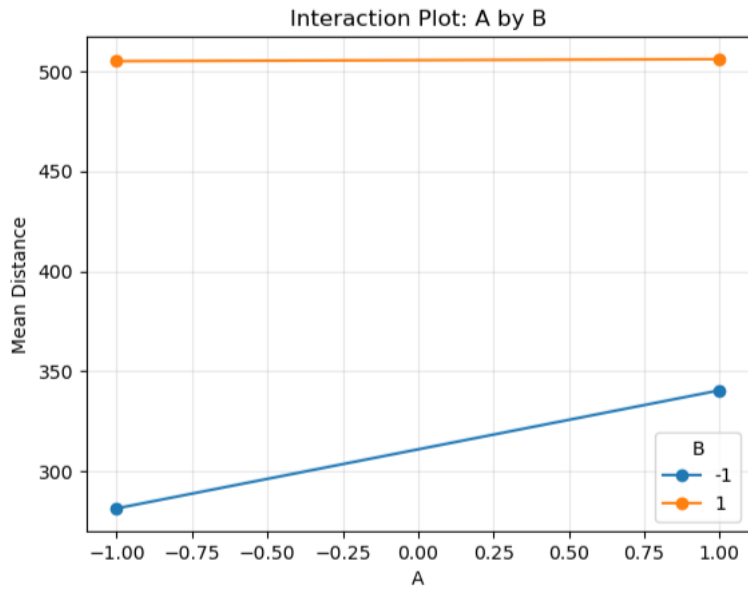


Figure 7: Interaction plot model of factor A by B not parallel

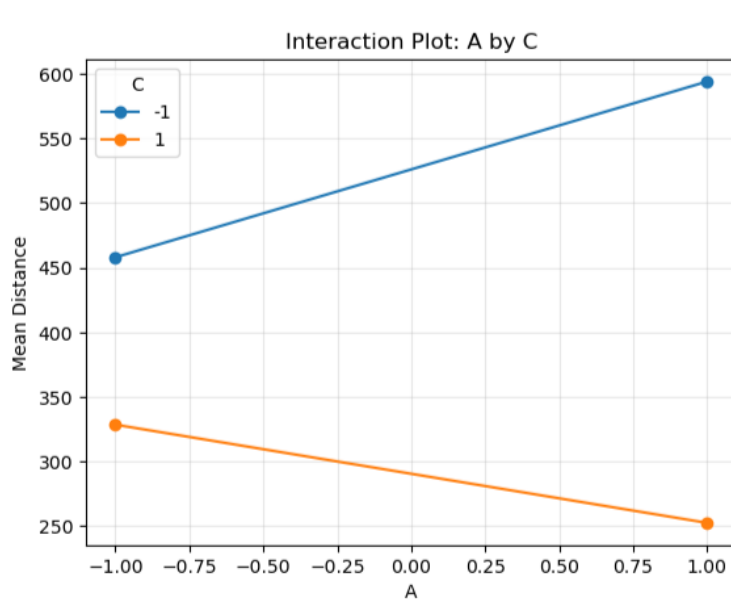


Figure 8: Interaction plot model of factor A by C not parallel

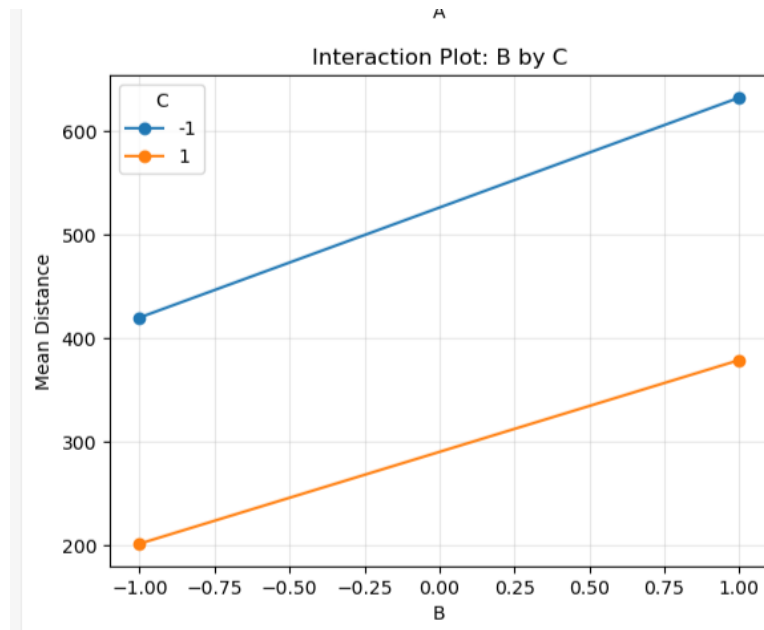


Figure 9: Interaction plot model of factor B by C being almost parallel

OLS Regression Results						
=====						
Dep. Variable:	Distance	R-squared:	0.869			
Model:	OLS	Adj. R-squared:	0.771			
Method:	Least Squares	F-statistic:	8.872			
Date:	Tue, 14 Apr 2026	Prob (F-statistic):	0.0306			
Time:	14:04:21	Log-Likelihood:	-44.012			
No. Observations:	8	AIC:	96.02			
Df Residuals:	4	BIC:	96.34			
Df Model:	3					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	408.2500	29.650	13.769	0.000	325.928	490.572
B	97.2500	29.650	3.280	0.031	14.928	179.572
C	-117.7500	29.650	-3.971	0.017	-200.072	-35.428
B:C	-8.7500	29.650	-0.295	0.783	-91.072	73.572
=====						
Omnibus:	3.495	Durbin-Watson:		3.004		
Prob(Omnibus):	0.174	Jarque-Bera (JB):		0.933		
Skew:	0.000	Prob(JB):		0.627		
Kurtosis:	1.327	Cond. No.		1.00		
=====						

Figure 10: OLS model of two main factors of B and C

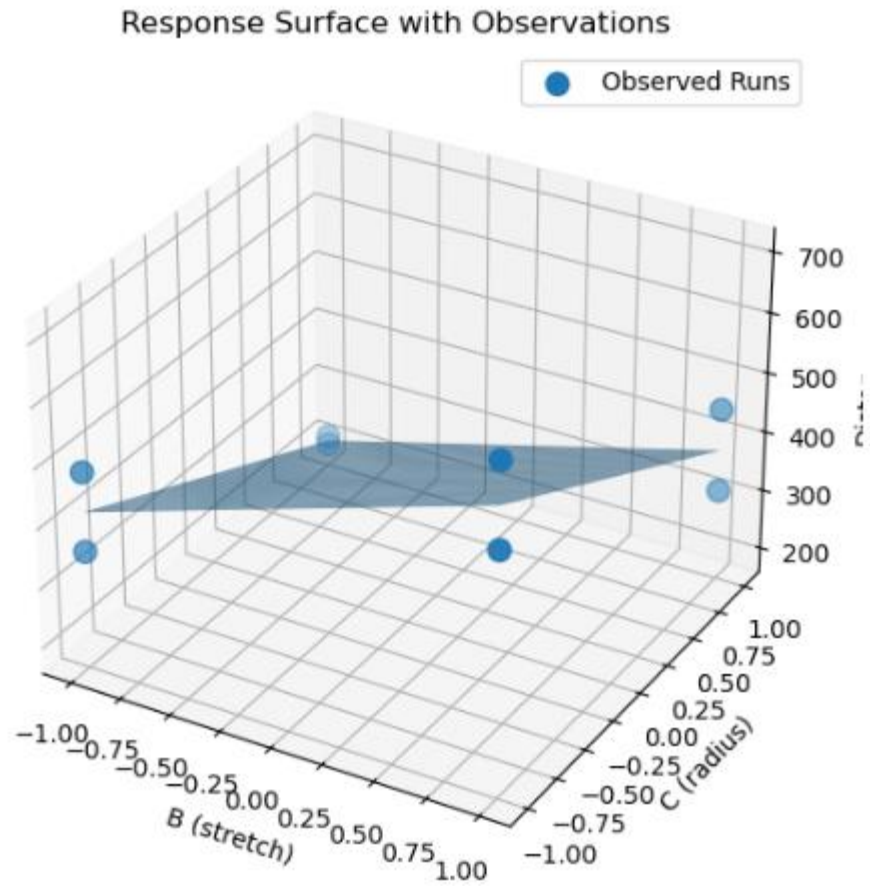


Figure 11: Surface model of two main factors of B and C in relation to distance

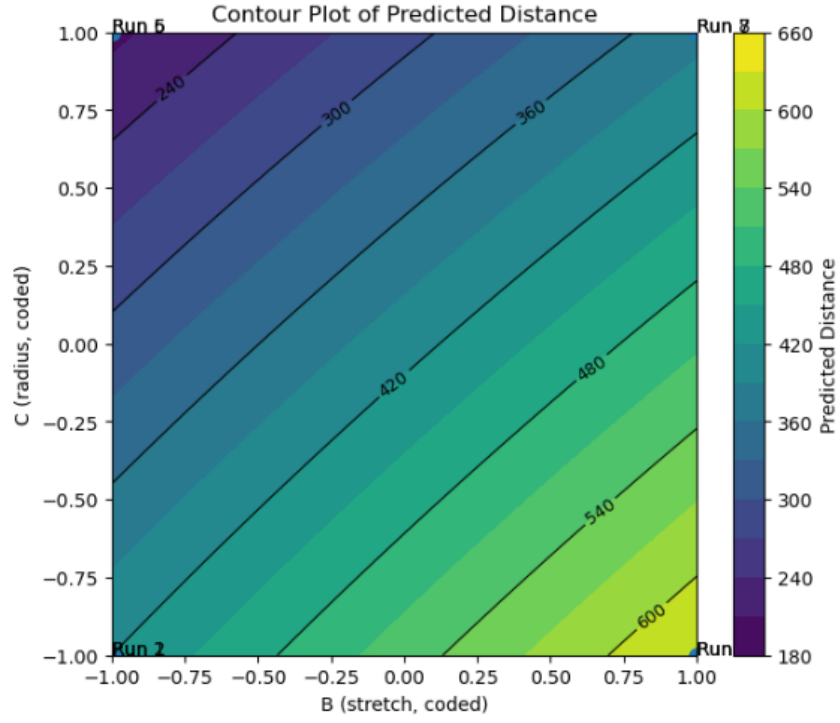


Figure 12: Contour model of two main factors of B and C in relation to distance

Table 3: Optimization steps of experiments of factor B and C in relation to max distance

Experiment	B (real world unit, cm)	C (real world unit, cm)	B (coded unit)	C (coded unit)	Y (distance)
Baseline	4.50	2.50	0	0	424.0
1	3.00	1.50	-1	-1	486.0
2	6.00	1.50	+1	-1	562.0
3	3.00	3.50	-1	+1	209.0
4	6.00	3.50	+1	+1	448.0
5	6.00	1.28	+1	-1.217	675.2
6	7.5	1.28	+2	-1.217	783.1